02/09/2026

CODE-A
Phase-1

Aakash
Medical | IIT-JEE | Foundations

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FORTNIGHTLY TEST FOR NEET-2027 (XII PASSED)

MM: 720 **FT-05** **Time: 180 Mins.**

Answers

1. (3)	37. (3)	73. (2)	109. (3)	145. (2)
2. (4)	38. (3)	74. (4)	110. (1)	146. (3)
3. (3)	39. (4)	75. (4)	111. (2)	147. (3)
4. (3)	40. (2)	76. (3)	112. (2)	148. (2)
5. (1)	41. (1)	77. (2)	113. (3)	149. (3)
6. (1)	42. (3)	78. (4)	114. (2)	150. (3)
7. (3)	43. (2)	79. (4)	115. (4)	151. (3)
8. (2)	44. (3)	80. (4)	116. (4)	152. (2)
9. (3)	45. (1)	81. (4)	117. (1)	153. (4)
10. (2)	46. (3)	82. (2)	118. (2)	154. (3)
11. (3)	47. (4)	83. (2)	119. (1)	155. (3)
12. (2)	48. (3)	84. (4)	120. (3)	156. (4)
13. (2)	49. (1)	85. (2)	121. (2)	157. (2)
14. (2)	50. (2)	86. (3)	122. (3)	158. (1)
15. (2)	51. (3)	87. (2)	123. (1)	159. (4)
16. (3)	52. (3)	88. (1)	124. (1)	160. (1)
17. (4)	53. (2)	89. (2)	125. (2)	161. (3)
18. (1)	54. (2)	90. (1)	126. (2)	162. (1)
19. (2)	55. (1)	91. (3)	127. (3)	163. (3)
20. (4)	56. (2)	92. (1)	128. (3)	164. (3)
21. (4)	57. (3)	93. (3)	129. (1)	165. (2)
22. (3)	58. (3)	94. (3)	130. (1)	166. (4)
23. (1)	59. (3)	95. (3)	131. (2)	167. (1)
24. (3)	60. (2)	96. (4)	132. (2)	168. (3)
25. (4)	61. (3)	97. (2)	133. (1)	169. (2)
26. (4)	62. (2)	98. (2)	134. (4)	170. (2)
27. (3)	63. (4)	99. (3)	135. (2)	171. (3)
28. (3)	64. (3)	100. (2)	136. (3)	172. (3)
29. (4)	65. (3)	101. (2)	137. (2)	173. (3)
30. (3)	66. (1)	102. (3)	138. (1)	174. (2)
31. (2)	67. (3)	103. (4)	139. (4)	175. (1)
32. (1)	68. (4)	104. (2)	140. (3)	176. (4)
33. (2)	69. (1)	105. (3)	141. (3)	177. (4)
34. (2)	70. (1)	106. (1)	142. (2)	178. (2)
35. (3)	71. (3)	107. (2)	143. (4)	179. (3)
36. (4)	72. (2)	108. (2)	144. (1)	180. (1)

Hints and Solutions

PHYSICS

(1) **Answer :** (3)
Solution:
$W_{isob} > W_{isot} > W_{adia}$ from area under *P-V* graph.

(2) **Answer :** (4)
Solution:
Natural convection requires gravity but forced convection can occur in gravity or gravity free space.

(3) **Answer :** (3)
Solution:
Here m_1 *is amount of water at 100 degree celcius and* m_2 *is amount of water at 10 degree celcius then*
$m_1 + m_2 = 20$ and Heat given by hotter water= heat absorbed by colder water,gives
$m_1 \times 65 = m_2 \times 25$
$m_1 \times 65 = (20 - m_1)25$
$m_1 = 5.6$ kg

(4) **Answer :** (3)
Solution:
For an isotropic solid ⇒ $\gamma = 3\alpha$

(5) **Answer :** (1)
Solution:
Solar constant, $S = KT^4$
$\left(\frac{\Delta S}{S} = \frac{4\Delta T}{T}\right) \times 100$
$\frac{\Delta S}{S} = 8\%$

(6) **Answer :** (1)
Solution:
As per Wien's Displacement Law,
λT = constant.
$\lambda 400 = \lambda' 1200$
$\lambda' = \frac{\lambda}{3}$
$\Delta\lambda = \frac{2\lambda}{3}$

(7) **Answer :** (3)
Solution:
For any thermometer
$\frac{\text{Reading} - \text{L.F.P}}{\text{U.F.P} - \text{L.F.P}}$ = Constant
$\frac{R-(-20)}{100-(-20)} = \frac{R-10}{90-10}$
$\Rightarrow \frac{R+20}{120} = \frac{R-10}{80}$
⇒ $8R + 160 = 12R - 120$
⇒ $R = 70°$

(8) **Answer :** (2)
Solution:
Heat loss = heat gain
$0.2 \times 1000 \times S \times (10 - 0) = 25 \times 80$
$200 \times S(10) = 2000$
$S = \frac{2000}{2000} = 1 \text{ cal g}^{-1} (\text{C}^\circ)^{-1}$

(9) **Answer :** (3)
Solution:
Let θ be temperature of interface, then in series heat current must be same.

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i.e. $\frac{\Delta\theta_1}{R_1} = \frac{\Delta\theta_2}{R_2}$, $R = \frac{l}{KA}$

$\Rightarrow \frac{2KA}{l}(100-\theta) = \frac{KA}{l}(\theta - 40)$

$\Rightarrow$ 200 – 2θ = θ – 40

240 = 3θ

$\Rightarrow \boxed{\theta = 80^\circ C}$

(10) Answer : (2)

Hint:

$P = e\sigma AT^4$

Solution:

$\frac{P_1}{P_2} = \frac{eA_1\sigma T_1^4}{eA_2\sigma T_2^4} = \frac{A_1}{A_2} \times \left(\frac{T_1}{T_2}\right)^4 = \left(\frac{r_1}{r_2}\right)^2 \times \left[\frac{T_1}{T_2}\right]^4$

$= \left[\frac{1}{2}\right]^2 \times \left[\frac{2000}{1000}\right]^4 = \frac{1}{4} \times \frac{2^4}{1^4} = \frac{4}{1} = 4$

(11) Answer : (3)

Solution:

$\frac{dT}{dt} = -k\left(T - T_0\right)$

$\int_{T_i}^{T_f} \frac{dT}{T-T_0} = -k\int_0^t dt \Rightarrow \ln\left(\frac{T_f - T_0}{T_i - T_0}\right) = -kt$

$\Rightarrow \left(T_f - T_0\right) = \left(T_i - T_0\right) \cdot e^{-kt}$

At $t = \infty$, $T_f = T_0$

(12) Answer : (2)

Solution:

$y = \alpha_1 + \alpha_2 + \alpha_3$

$= (4 + 5 + 1) \times 10^{-2}/^\circ C$

$= 10 \times 10^{-2}/^\circ C$

(13) Answer : (2)

Solution:

$\frac{T_1 - T_2}{\Delta t} = K\left(\frac{T_1 + T_2}{2} - T_0\right)$

$\frac{3T - 2T}{10} = K(2.5T - T)$

$\Rightarrow \frac{T}{10} = K(1.5)T$

$K = \frac{1}{15}$

Now, $\frac{T - x}{10} = K\left(\frac{T + x}{2} - T\right)$

Solving $x = \frac{3T}{2}$

(14) Answer : (2)

Solution:

Thermal strain = Longitudinal strain = $\alpha \Delta T$

$\Rightarrow$ Longitudinal strain, $\delta = 10^{-5} \times 10^2 = 10^{-3}$

$\Rightarrow$ Compressive stress = δ × Young's Modulus

$= 10^{-3} \times 0.5 \times 10^{11}$

$= 0.5 \times 10^8$

$\Rightarrow$ Compressive force = $0.5 \times 10^8 \times 10^{-3} = 0.5 \times 10^5$

$= 5 \times 10^4$

$= 50 \times 10^3$ N

(15) Answer : (2)

Solution:

$t + r + a = 1$

$a = 1 - (t + r)$

$= 1 - \left(\frac{1}{9} + \frac{1}{6}\right)$

$a = \frac{13}{18}$

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(16) **Answer :** (3)
Solution:

a. Isothermal (iv) Temperature constant
b. Isobaric (i) Pressure constant
c. Isochoric (ii) Volume constant
d. Adiabatic (iii) No heat exchange between system and surrounding

(17) **Answer :** (4)
Solution:
$\Delta Q = \Delta U + W$
When heat is added to a gaseous system, the temperature of the gas may increase, but it is not guaranteed. The temperature change depends on the process the gas undergoes. For example, in an isothermal process, the temperature remains constant even when heat is added, as the gas does work on the surroundings. Therefore, the correct statement is that the temperature may increase or remain unchanged. Hence, the correct option is (4) i.e Both (2) & (3).

(18) **Answer :** (1)
Solution:
For an adiabatic process, $\Delta Q = 0$
$\Rightarrow \Delta U + \Delta W = 0$
$\Rightarrow \Delta U = -\Delta W$
For an isothermal process, $\Delta T = 0$
$\Rightarrow \Delta U = 0$

(19) **Answer :** (2)
Solution:
$P \propto V$
PV^{-1} = Constant
So, N = – 1
Then
$C = \frac{R}{\gamma-1} + \frac{R}{1-N}$
$= \frac{R}{\left(\frac{5}{3}-1\right)} + \frac{R}{1-(-1)}$
$= \frac{3R}{2} + \frac{R}{2} = 2R$

(20) **Answer :** (4)
Solution:
$Q = W + \Delta U$
$Q = x\Delta U + \Delta U$
$Q = (x+1)\Delta U$
$nC_p\Delta T = (x+1)nC_v\Delta T$
$\Rightarrow \quad \gamma = x+1$
$\frac{5}{3} = x+1$
$x = \frac{2}{3}$

(21) **Answer :** (4)
Solution:
$Q = \Delta U + W$
$0 = -75 + W$
$W = 75$ J

(22) **Answer :** (3)
Solution:
$\left.\begin{matrix} P^2V = \text{constant} = K \\ PV = nRT \end{matrix}\right\}$ hence $\sqrt{V} = \frac{nRT}{K}$
$\sqrt{\frac{V}{2V}} = \frac{T}{T'}$
$\Rightarrow T' = \sqrt{2}T$

(23) **Answer :** (1)
Solution:
In adiabatic process heat exchange is zero

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(24) **Answer :** (3)
Solution:
in isochoric process $W = 0$
First law of thermodynamics $\Delta Q = \Delta U + W$

(25) **Answer :** (4)
Hint:
$\Delta Q = nC_p \Delta T$
Solution:
$\Delta Q = nC_p \Delta T$
$= 5\left(1+\frac{5}{2}\right)R \times 30$
$= 5\left(\frac{7}{2}\right)2 \times 30$
ΔQ = 1050 cal

(26) **Answer :** (4)
Solution:
$PT^{\frac{1}{1-\gamma}}$ = constant not an adiabatic process.

(27) **Answer :** (3)
Solution:
For an isochoric process, volume remains constant

(28) **Answer :** (3)
Solution:
$W_{A\to B} = P_0V_0, W_{B\to c} = 0$
$W_{c\to D} = -4P_0V_0$
$W_{net} = -3P_0V_0$

(29) **Answer :** (4)
Solution:
$C_p - C_V = R$
$\therefore\ C_v = C_p - R = \frac{5R}{2} - R = \frac{3R}{2}$

(30) **Answer :** (3)
Solution:
Answer (3)
$\frac{\Delta Q}{\Delta U} = \frac{nC_p\Delta T}{nC_v\Delta T} = \frac{C_p}{C_v} = \gamma$

(31) **Answer :** (2)
Solution:
$v_{\text{rms}} = \sqrt{\frac{3RT}{M}},\ P = \frac{1}{3}\rho(v_{\text{rms}})^2,$
$U = nC_VT$, Translational $K.E. = \frac{3}{2}K_BT$

(32) **Answer :** (1)
Hint:
Use, $\frac{C_P}{C_V} = \gamma$
Solution:
$\gamma = \frac{f+2}{f}$
Degree of freedom for a monatomic gas is 3, and for a diatomic gas is 5.
$\gamma_{\text{mono}} = \frac{5}{3}$
$\gamma_{\text{dia}} = \frac{7}{5}$

(33) **Answer :** (2)
Solution:
Total kinetic energy of a gas depends on its temperature.
Gas molecules move in random direction with different velocities.

(34) **Answer :** (2)
Solution:

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The mean free path is given by

$\lambda = \frac{1}{\sqrt{2}\pi d^2 n}$

$n = \frac{N}{V}$

$\lambda = \frac{1}{\sqrt{2}\pi d^2 \frac{N}{V}} = \frac{V}{\sqrt{2}\pi d^2 N}$

$PV = nRT$

$\lambda = \frac{nRT}{\sqrt{2}\pi d^2 NP}$

$\lambda \propto T, \lambda \propto V$

$\lambda \propto \frac{1}{P}$

(35) Answer : (3)
Solution:
PT = constant
$PV^{1/2}$ = constant
$C = C_V + \frac{R}{1-x}$
$\frac{7R}{2} = \frac{f}{2}R + \frac{R}{1-\frac{1}{2}}$
$\frac{f}{2}R = \frac{3}{2}R$
f = 3

(36) Answer : (4)
Solution:
PV = *nRT*
$P = nRT\left(\frac{1}{V}\right)$

(37) Answer : (3)
Solution:
f_m = $\frac{n_1 f_1 + n_2 f_2}{n_1 + n_2}$ [f_m → degree of freedom of mixture]
f_m = $\frac{3+5}{1+1}$ = 4
$\because \gamma_m = 1 + \frac{2}{f_m} = \frac{3}{2}$

(38) Answer : (3)
Hint:
Thermal capacity = *m* × specific heat
Solution:
$50 \times 0.2 = 10$ cal/°C

(39) Answer : (4)
Solution:
$\because V_{rms}$ = $\sqrt{\frac{RT}{M}}$ *M* → Molar mass
$\therefore \frac{V_1}{V_2} = \sqrt{\frac{M_2}{M_1}}$ $V_1 \rightarrow For\ helium$ $V_2 \rightarrow For\ neon$
$\frac{V_1}{V_1} = \sqrt{\frac{22}{4}}$ = 2.3

(40) Answer : (2)
Solution:
The force of interaction between molecules is assumed to be negligible according to kinetic theory of gases.

(41) Answer : (1)
Solution:
$E = \frac{3}{2} k_B T$
$\frac{E_1}{E_2} = \frac{T_1}{T_2} = \frac{(273+27)}{(273+227)} = \frac{3}{5}$
E_2 = $\frac{5}{3} \times E_1 = \frac{5}{3} \times 6.2 \times 10^{-21}$

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= 10.333 × 10^{-21} J

(42) **Answer :** (3)
Solution:
$V_{av} = \sqrt{\frac{8RT}{\pi M}}$
$V_{mP} = \sqrt{\frac{2RT}{M}}$
$\frac{V_{av}}{V_{mP}} = \frac{2}{\sqrt{\pi}}$

(43) **Answer :** (2)
Solution:
CO molecule is diatomic. At high temperature, it will be having 7 degrees of freedom, as the vibrational mode is also excited.

(44) **Answer :** (3)
Solution:
$u_1 = u_2$
$f_1 n_1 T_1 = f_2 n_2 T_2$
$\frac{n_1}{n_2} = \frac{f_2 T_2}{f_1 T_1} = \frac{(3)}{(5)} \frac{(2T)}{T} = \frac{6}{5}$

(45) **Answer :** (1)
Solution:
A real gas behaves like an ideal gas at high temperature and low pressure.

CHEMISTRY

(46) **Answer :** (3)
Solution:
Physical equilibrium are dynamic in nature.
Both the opposing processes occur at the same rate during equilibrium.

(47) **Answer :** (4)
Solution:
The equilibrium constant does not depend upon initial concentrations and has a definite value for every chemical reaction at a given temperature.

(48) **Answer :** (3)
Solution:
$\mathrm{FeO(s) + CO(g) \rightleftharpoons Fe(s) + CO_2(\,g)}$
$\mathrm{K_p = \frac{P_{CO_2}}{P_{CO}} = \frac{x}{3-x} = 0.5}$
2x = 3 – x ⇒ x = 1 atm

(49) **Answer :** (1)
Hint:
Equilibrium constant for the reverse reaction is the inverse of the equilibrium constant for the reaction in forward direction.
Solution:
$\mathrm{N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g),\ K_c}$ = 16 ...(1)
On reversing the above chemical reaction,
$\mathrm{2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g),\ K_c' = \frac{1}{K_c}}$ ...(2)
Now, divide equation 2 by 2
$\mathrm{NH_3(g) \rightleftharpoons \frac{1}{2} N_2(g) + \frac{3}{2} H_2(g),\ K_c(new) = \sqrt{\frac{1}{K_c}}}$
$\mathrm{K_c(new) = \sqrt{\frac{1}{16}} = \frac{1}{4} = 0.25}$

(50) **Answer :** (2)
Solution:
$\mathrm{K_P = K_C(RT)^{\Delta ng}}$
$\mathrm{\frac{K_C}{K_P} = \frac{1}{(RT)^{\Delta ng}}}$

$\frac{K_C}{K_P} = \left(RT\right)^{-\Delta ng}$

a. $\Delta n_g = 2$

$\frac{K_c}{K_p} = (RT)^{-2}$

$\frac{K_c}{K_p} = \frac{1}{(RT)^2}$

b. $\Delta n_g = 0$

$\frac{K_c}{K_p} = \left(RT\right)^0 = 1$

c. $\Delta n_g = 1$

$\frac{K_c}{K_p} = \left(RT\right)^{-1} = \frac{1}{RT}$

d. $\Delta n_g = -1$

$\frac{K_c}{K_p} = \left(RT\right)^{-\Delta n_g}$

$= (RT)^1$

(51) Answer : (3)
Solution:
In homogeneous equilibrium, all the reactants and products are in same phase.

(52) Answer : (3)
Solution:
With change of partial pressure.

...[1 Mark]

K_P remain constant.

P(s) $\rightleftharpoons$2Q(g) + 3R(g)

$K_P = (P_Q)^2 (P_R)^3 = (P'_Q)^2 \ (3P_R)^3$

$P_Q^2 \ P_R^3 = {P'_Q}^2 \times 27 P_R^3$

$\frac{P_Q^2}{27} = {P'_Q}^2$

$P'_Q = \frac{P_Q}{3\sqrt{3}}$

(53) Answer : (2)
Solution:

Initial molar concentration $\underset{6}{A_2} + \underset{5}{B_2} \rightleftharpoons 2\underset{2.5}{AB}$

Equ. molar concentration $\underset{5M}{(6-x)} \quad \underset{4M}{(5-x)} \quad \underset{4.5M}{(2.5+2x)}$

2.5 + 2x = 4.5
2x = 2
x = 1

$K_C = \frac{(AB)^2}{(A_2)(B_2)} = \frac{(4.5)^2}{5\times4} = \frac{4.5\times4.5}{5\times4} = 1.01$

(54) Answer : (2)

(55) Answer : (1)
Hint:

$NaF \rightleftharpoons \underset{(S)}{Na^+} + \underset{(S)}{F^-}$

(S = 0.1 (given))

$CaF_2 \rightleftharpoons \underset{(S)}{Ca^{2+}} + \underset{(2S+0.1)}{2F^-}$

(Since S < < < 0.1)
(∴ 2S + 0.1 = 0.1)
Solution:

K_{sp} = (S) $(0.1)^2$

$\frac{7.2\times10^{-9}}{(0.1)^2} = (S)$

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$7.2 \times 10^{-7} = S$

(56) **Answer :** (2)
Solution:
Addition of helium gas at constant pressure in the above reaction will shift the reaction in a direction where the number of moles of gases increase

(57) **Answer :** (3)
Solution:
Aqueous solution of salt of strong acid and strong base is neutral.

(58) **Answer :** (3)
Solution:

When $OH^-_{SB} < 10^{-7}\ M$, contribution of OH^- from H_2O is also considered

$$H_2O \rightleftharpoons \underset{x}{H^+(aq)} + \underset{10^{-8}+x}{OH^-(aq)}$$

$$10^{-14} = x\left(10^{-8} + x\right)$$

Solving quadratic,

$[H^+] = x = 9.5 \times 10^{-8}$

pH = 7.02

(59) **Answer :** (3)

(60) **Answer :** (2)
Solution:
$\Delta G° = -2.303\ RT\log K$

$= -2.303 \times 300 \times R \log 10^2$

$\Delta G° = -2.303 \times 600 \times R$

(61) **Answer :** (3)
Solution:

	CH_3COOH	+ $NaOH$ →	CH_3COONa	+ H_2O
Initially milli moles	$0.2 \times 250 = 50$	$0.1 \times 200 = 20$	×	×
After reaction milli moles	$50 - 20 = 30$	$20 - 20 = 0$	20	20

Finally solution contains weak acid and its salt with strong base, so it is a case of acidic buffer.

$$pH = pK_a + \log\frac{[Salt]}{[Acid]}$$

$$= 4.74 + \log\left[\frac{20}{30}\right]$$

$= 4.74 + \log 2 - \log 3$

$= 4.74 + 0.30 - 0.48$

$= 4.56$

(62) **Answer :** (2)
Solution:

$K_a \times K_b = K_w = 10^{-14}$

$K_b = \frac{10^{-14}}{K_a} = \frac{10^{-14}}{10^{-6}} \Rightarrow 10^{-8}$

$\log K_b = -8$

$pK_b = -\log K_b = 8$

(63) **Answer :** (4)
Solution:
NH_3 is a Lewis base

(64) **Answer :** (3)
Solution:
$V_1 = 100$ ml $V_2 = 1000$ ml

$M_1 = 10^{-2}$ M_2 = ?

$M_2 = \frac{10^{-2} \times 100}{1000} = 10^{-3}$ M

pH = – log (10^{-3}) = 3

(65) **Answer :** (3)

(66) **Answer :** (1)
Solution:

$H_2A \rightleftharpoons H^+ + HA^-$; K_{a_1}

$HA^- \rightleftharpoons H^+ + A^{2-}$; K_{a_2}

$K_a = K_{a_1} \times K_{a_2}$

(67) **Answer :** (3)
Solution:

CH_3COOH +	$NaOH$ →	$CH_3COONa + H_2O$	
$initial$	40 $mmol$	10 $mmol$	0
$final$	30 $mmol$	0	10 $mmol$

Hence, buffer solution

(68) **Answer :** (4)
Solution:

$\left[H^+\right] = \frac{N_1V_1 + N_2V_2 + \ldots}{V_1 + V_2 \ldots}$

Sol.: If pH = 2 then $N_1 = 10^{-2}$

pH = 3 then $N_2 = 10^{-3}$

$\left[H^+\right] = \frac{N_1V + N_2V}{2V}$

$= \frac{10^{-2} \times V + 10^{-3}V}{2V} = \frac{1.1 \times 10^{-2}}{2} = 5.5 \times 10^{-3}$ M

$pH = -\log\left[H^+\right] = -\log\left(5.5 \times 10^{-3}\right) = 2.26$

$\simeq 2.3$

(69) **Answer :** (1)
Solution:
Catalyst decreases the activation energy for the forward and reverse reaction.

(70) **Answer :** (1)
Solution:
When CH_3COONa is added to CH_3COOH solution, degree of dissociation decreases due to common ion effect/

(71) **Answer :** (3)
Solution:
Solubility of $CuSO_4$ is maximum in NH_3

$CuSO_4 + 4NH_3 \rightarrow \left[Cu(NH_3)_4\right]SO_4$

(72) **Answer :** (2)
Solution:

$pOH = pK_b + \log \frac{[\text{salt}]}{[\text{base}]} = 5 + \log \frac{10}{1} = 6$

(73) **Answer :** (2)
Solution:

$K_P = K_C(RT)^{\Delta n_g}$

For $K_P = K_C$ ($\Delta n_g = 0$)

$K_P \neq K_C$ ($\Delta n_g \neq 0$)

(1) $H_2O(g) + CO(g) \rightleftharpoons H_2(g) + CO_2(g)$

$\Delta n_g = 0$ ($K_P = K_C$)

(2) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

$\Delta n_g = -2$ ($K_P \neq K_C$)

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(3) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
$\Delta n_g = 0$ ($K_P = K_C$)
(4) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$
$\Delta n_g = 0$ ($K_P = K_C$)

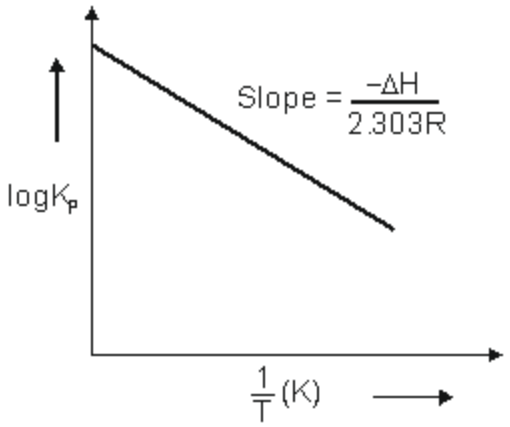
(74) Answer : (4)
Solution:
$CaCO_3(s) \rightleftharpoons Ca^{+2} + CO_3^{-2}$
Precipitation take place when
Ionic product > K_{SP}
$[Ca^{+2}]\ [CO_3^{-2}] > K_{SP}$
For (3) option
$\frac{10^{-4}}{2} \times \frac{10^{-3}}{2} > 4.7 \times 10^{-9}$
$\frac{1}{4} \times 10^{-7} > 4.7 \times 10^{-9}$
$\therefore$ Precipitation take place.

(75) Answer : (4)
Solution:
Endothermic reactions are favoured at high temperature and exothermic reactions are favoured at low temperature.

(76) Answer : (3)
Solution:
$BaSO_4$: $s^2 = K_{sp}$
$\therefore\ s = \sqrt{K_{sp}}$
Hg_2Cl_2 : $s \times (2s)^2 = K_{sp}$
$= 4s^3 = K_{sp}$
$s = \sqrt[3]{\frac{K_{sp}}{4}} = \left(\frac{K_{sp}}{4}\right)^{1/3}$
$Ca_3(PO_4)_2$: $(3s)^3 \times (2s)^2 = K_{sp}$
$\Rightarrow 108\ s^5 = K_{sp}$
$s = \sqrt[5]{\frac{K_{sp}}{108}}$
$Fe(OH)_3$: $s \times (3s)^3 = K_{sp}$
$\Rightarrow 27s^4 = K_{sp}$
$\Rightarrow s = \sqrt[4]{\frac{K_{sp}}{27}}$

(77) Answer : (2)
Hint:
$\log K_P = \log A - \frac{\Delta H}{2.303\ RT}$
Solution:
For an endothermic reaction, $\Delta H > 0$

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(78) **Answer :** (4)
Solution:
If $\Delta n_g = 0$ there is no effect of the change of pressure.

(79) **Answer :** (4)
Hint:
Salt of weak acid and weak base undergo both cationic and anionic hydrolysis
Solution:
1. $(NH_4)_2SO_4$ – Salt of weak base and strong acid so, NH_4^+ will undergo cationic hydrolysis

2. CH_3COONa – Salt of weak acid and strong base so CH_3COO^- will undergo anionic hydrolysis.

3. CH_3COONH_4 – Salt of weak acid and weak base, so, both CH_3COO^- and NH_4^+ will undergo hydrolysis i.e., anionic and cationic hydrolysis.
4. Na_2SO_4 – Salt of strong acid and strong base and it will not undergo hydrolysis.

(80) **Answer :** (4)
Solution:

$$K_C = \frac{[AB]^2}{[A_2][B_2]}$$

$$= \frac{(4\times10^{-2})^2}{(1\times10^{-2})(2\times10^{-2})} = 8$$

(81) **Answer :** (4)
Solution:

$$A_xB_y \rightleftharpoons \underset{xs}{xA^{y+}} + \underset{ys}{YB^{x-}}$$

$K_{sp} = [A^{y+}]^x [B^{x-}]^y$
$= (xs)^x (ys)^y$
$= x^x y^y (s)^{x+y}$

(82) **Answer :** (2)
Hint:
Apply Le Chatelier's principle.
Solution:
· Low temperature favours forward reaction as it is exothermic.
· High pressure shifts the equilibrium towards lesser number of gaseous mole *i.e.*, forward direction.

(83) **Answer :** (2)
Solution:
$K_{sp}[AgI] = [Ag^+][I^-]$
$8.5 \times 10^{-17} = 0.1\ [I^-]$
$[I^-] = 8.5 \times 10^{-16}$ M
$K_{sp}[Hg_2I_2] = [Hg_2^{2+}][I^-]^2$
$2.5 \times 10^{-26} = 0.1[I^{-1}]^2$
$[I^-] = 5.0 \times 10^{-13}$ M

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[I^-] to precipitate AgI is smaller
∴AgI will start precipitating first
On further addition of I^-more AgI will precipitate
And when $[I^-] \geq 5 \times 10^{-13} M$, Hg_2I_2 will start precipitating.

(84) Answer : (4)
Hint:
On adding equations, equilibrium constants get multiplied.
Solution:

A $\rightleftharpoons$ B; $K_c = 10$

B $\rightleftharpoons$ C; $K_c = 5$

C $\rightleftharpoons$ D; $K_c = 2$

A $\rightleftharpoons$ D; $K_c = 10 \times 5 \times 2 = 100$

(85) Answer : (2)
Solution:

$X^-(aq) + H_2O(l) \overset{K_h}{\rightleftharpoons} HX(aq) + OH^-(aq)$

$K_h = \frac{K_w}{K_a} = \frac{10^{-14}}{10^{-5}} = 10^{-9}$

(86) Answer : (3)
Solution:

$\mathrm{pOH} = \mathrm{pK_b} + \log \frac{[\mathrm{NH_4^+}]}{[\mathrm{NH_4\,OH}]}$

$4.7 = \mathrm{pK_b} + \log\left(\frac{0.2}{0.2}\right)$

⇒ $pK_b = 4.7$

(87) Answer : (2)
Solution:

$\mathrm{pH} = 5 \Rightarrow \left(\mathrm{H^+}\right) = 10^{-5}\mathrm{M}$

1000 ml contains 10^{-5} mol
∴ 10 ml will contains 10^{-7} mol
No. of H^+ ions = $10^{-7} \times 6.022 \times 10^{23}$
$= 6.022 \times 10^{16}$

(88) Answer : (1)
Solution:

	$\mathrm{P(g)}$	+	$\mathrm{Q(g)}$	$\rightleftharpoons$	$\mathrm{2R(g)}$
$\mathrm{t=0}$	5		2		0
$\mathrm{t=t_q}$	$5-\mathrm{x}$		$2-\mathrm{x}$		$2\mathrm{x}$

$n_R = 0.5$ mol
$2x = 0.5$
∴ At equilibrium $x = 0.25$

$\therefore \left[\mathrm{P}\right] = \frac{5-\mathrm{x}}{1\mathrm{L}} = \frac{5-0.25}{1} = 4.75\ \mathrm{M}$

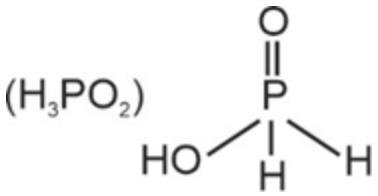
$\left[\mathrm{Q}\right] = \frac{2-\mathrm{x}}{1\mathrm{L}} = 1.75\ \mathrm{M}$

$\left[\mathrm{R}\right] = \frac{0.5}{1} = 0.5\ \mathrm{M}$

$\therefore \quad \mathrm{K_{eq}} = \frac{[\mathrm{R}]^2}{\left[\mathrm{P}\right]\left[\mathrm{Q}\right]} = \frac{(0.5)^2}{4.75 \times 1.75} = 0.03$

(89) Answer : (2)
Solution:

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$H_2PO_2^-$ does not have acidic hydrogen so it does not form conjugate base

Conjugate base of $H_2PO_4^-$ is HPO_4^{2-}

Conjugate base of HPO_4^{2-} is PO_4^{3-}

(90) Answer : (1)

BOTANY

(91) Answer : (3)
Solution:
In roots, cuticle and hypodermis are absent.
The outermost layer of stele is pericycle. In roots, endodermis has depositions of suberin in the form of casparian strips.

(92) Answer : (1)
Solution:
B represents pericycle which is in the form of semilunar patches comprise of sclerenchymatous cell.
C represents cambium, meristematic tissue.
D represents metaxylem (later formed primary xylem).

(93) Answer : (3)
Solution:
Bulliform cell-monocot leaf
Epiblema-root

(94) Answer : (3)
Solution:
i) Monocotyledonous roots have more vascular bundles than dicotyledonous roots.
ii) Cortex below hypodermis of the stem consists of thin-walled parenchyma cells.

(95) Answer : (3)
Solution:
The trichomes in the shoot system are usually multicellular. They may be branched or unbranched and soft or stiff. They may even be secretory. The trichomes help in preventing water loss due to transpiration.

(96) Answer : (4)
Solution:
Endodermis is absent in monocot stem.

(97) Answer : (2)
Hint:
In monocot stems, vascular bundles are scattered.
Solution:
A large number of vascular bundles are arranged in a ring; the 'ring' arrangement of vascular bundles is a characteristic of a dicot stem.

(98) Answer : (2)
Solution:
In dicot roots, vascular bundles is diarch to hexarch. Vascular bundles is polyarch in monocot root.

(99) Answer : (3)
Solution:
The monocot stem has a large number of conjoint, closed and scattered vascular bundles.

(100) Answer : (2)
Solution:
The layer of cortex in dicot roots that is characterised by the presence of casparian strips is endodermis. It is comprised of a single layer of barrel-shaped cells without any intercellular spaces.

(101) Answer : (2)

Solution:
In conjoint type of vascular bundles, the xylem and phloem are situated at the same radius of vascular bundles. Such vascular bundles are common in stems and leaves

(102) Answer : (3)
Solution:
In dicot leaves, mesophyll is divided into palisade mesophyll and spongy mesophyll.

(103) Answer : (4)
Solution:
In dicotyledonous stem, the cells in the innermost layer of the cortex called the endodermis are rich in starch grains and this layer is also referred to as the starch sheath.

(104) Answer : (2)
Solution:
The given figure is of radial vascular bundles.
Radial vascular bundles are found in roots of both monocots and dicot roots.

(105) Answer : (3)
Hint:
Epidermis remains in direct contact with the external environment.
Solution:
Epidermis forms a continuous layer and it is generally single-layered.

(106) Answer : (1)
Hint:
Isobilateral leaves have nearly similar sized vascular bundle.
Solution:
Parallel venation in monocot leaves is reflected in the near similar sizes of vascular bundles.

(107) Answer : (2)
Solution:
Formation of root hairs is a function of epidermal tissue system.

(108) Answer : (2)
Solution:
Monocot stem has epidermis and hypodermis but not endodermis, pericycle and pith.

(109) Answer : (3)
Solution:
Ground tissue system forms the main bulk of plant and is divided into three zones. The vascular bundle form the conducting tissue.

(110) Answer : (1)
Solution:
The above given figure is of monocot stem.

(111) Answer : (2)
Solution:
Conjunctive tissue cannot be photosynthetic as they are found in roots.

(112) Answer : (2)
Solution:
Guard cells are parenchymatous.

(113) Answer : (3)
Hint:
All tissues on the inner side of endodermis constitute stele.
Solution:
Endodermis is innermost layer of cortex, and present outside the pericycle. It is not the part of stele.

(114) Answer : (2)
Solution:
In monocot stem, hypodermis is sclerenchymatous.

(115) Answer : (4)
Solution:
In monocot leaf, ground tissue system, *i.e.,* mesophyll is not differentiated.

(116) Answer : (4)
Solution:

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Stomatal apparatus consist of stomatal pore, guard cells and surrounding subsidiary cell.

(117) Answer : (1)
Hint:
Underground roots do not have collenchymatous tissue.
Solution:
Hypodermis is absent in roots.

(118) Answer : (2)
Solution:
Conjunctive tissue is made up of parenchymatous cells lie between the xylem and the phloem in roots.

(119) Answer : (1)
Solution:
A few epidermal cells, in the vicinity of the guard cells modifies into subsidiary cells.

(120) Answer : (3)
Solution:
Phloem parenchyma is usually absent in the monocot plants.

(121) Answer : (2)
Hint:
Isobilateral leaves have equal number of stomata on both the surfaces.
Solution:
Isobilateral leaves are found in monocots. They lack palisade parenchyma & have close type of vascular bundles.

(122) Answer : (3)
Solution:
The primary adaptive advantage provided by bulliform cells in grasses during water stress is that they reduce transpiration by causing leaf rolling to minimize exposed surface area.

(123) Answer : (1)
Solution:
Dumb-bell shaped guard cells are seen in grasses.

(124) Answer : (1)
Solution:
When the bulliform cells in the leaves have absorbed water and are turgid, the leaf surface is exposed. When they are flaccid due to water stress, they make the leaves curl inwards to minimise water loss

(125) Answer : (2)
Hint:
Vascular tissue system consists of complex tissues, the phloem and xylem.
Solution:
Ground tissue system in plants is considered as the fundamental tissue system.

(126) Answer : (2)
Solution:
The size of vascular bundles in a dorsiventral leaf is dependent on size of veins.

(127) Answer : (3)
Solution:
Semi-lunar patches of sclerenchymatous pericycle is a feature of dicot stem.

(128) Answer : (3)
Solution:
Pith or medulla is present at centre of dicot stem.

(129) Answer : (1)
Solution:
The epidermal tissue system comprises of epidermal cells, stomata and the epidermal appendages.

(130) Answer : (1)
Solution:
Vascular bundles without cambium are closed type.

(131) Answer : (2)
Solution:
Cells may or may not be similar structurally.

(132) Answer : (2)

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Solution:
Endodermis has casparian strips.They are impervious to water molecule

(133) Answer : (1)
Solution:
Vessels are the elements of xylem.

(134) Answer : (4)
Solution:
Dicot leaves have palisade parenchyma below the adaxial epidermis.

(135) Answer : (2)
Solution:
When xylem and phloem are arranged in an alternate manner along different radii, it is called radial V.B. as seen in roots.

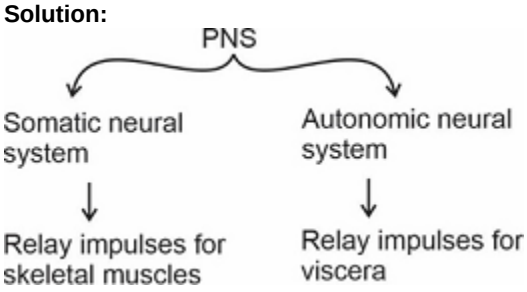
ZOOLOGY

(136) Answer : (3)
Solution:
Saddle joint is present between carpal and metacarpal of thumb

(137) Answer : (2)
Solution:
Dura mater → Arachnoid → Pia mater is arranged from outside to inside.

(138) Answer : (1)
Solution:
Thoracic vertebrae, ribs and sternum together form the rib cage.

(139) Answer : (4)
Solution:



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(140) Answer : (3)
Solution:
Transportation of substances is function of circulatory system of the body. It is not the function of skeletal system.

(141) Answer : (3)
Solution:
Scapula is a large triangular flat bone situated in the dorsal part of the thorax between the second and the seventh ribs.

(142) Answer : (2)
Solution:
Gout: Inflammation of joints due to accumulation of uric acid crystals.

(143) Answer : (4)
Solution:
Saddle joint is a synovial joint.

(144) Answer : (1)
Solution:
Osteoporosis is an age-related disorder characterised by decreased bone mass and increased chances of fractures.

(145) Answer : (2)
Solution:
Neural system of insects consists of a brain, neural tissues with a number of ganglia.

(146) Answer : (3)
Solution:

ECF contains 30 times less K^+ than axoplasm while ECF contains 10 times more Na^+ than ICF. Potential difference around -70 mV exists across axolemma due to differential distribution of ions. Neural membrane is impermeable to movement of negatively charged proteins. The exterior surface of axolemma is positively charged.

(147) Answer : (3)
Solution:
During depolarization, the outer surface of the neural membrane ultimately becomes negatively charged as compared to the inner surface and leads to action potential or nerve impulse.

(148) Answer : (2)
Solution:
Bipolar neurons have one axon and one dendrite.

(149) Answer : (3)
Solution:
In electrical synapses, the communication between neurons occurs through gap junctions, not tight junctions. Gap junctions allow direct passage of ions and small molecules between neurons, facilitating rapid transmission of impulses.

(150) Answer : (3)
Solution:
The two halves of pelvic girdle meet ventrally to form pubic symphysis containing fibrous cartilage.

(151) Answer : (3)
Solution:
The PNS comprises of all the nerves of the body (both cranial and spinal nerves)

(152) Answer : (2)
Solution:
Repolarisation results in positive charge on the outer side and negative charge on the inner side of axolemma.

(153) Answer : (4)
Solution:
Cerebellum of hindbrain is concerned with regulating body posture, equilibrium and coordinated rapid muscular activity.

(154) Answer : (3)
Solution:
Dendrites receive impulses from other neurons and transmit these impulses towards the cell body of the neuron.

(155) Answer : (3)
Solution:
Cranial bones are 8 in number.

(156) Answer : (4)
Solution:
In chemical synapse, the membranes of the pre- and post-synaptic neurons are separated by a fluid-filled space called synaptic cleft.

(157) Answer : (2)
Solution:
Patella, parietal and scapula are paired bones while hyoid is an unpaired bone present at the base of buccal cavity.

(158) Answer : (1)
Solution:
Associated deep structures like amygdala, hippocampus *etc.* form a complex structure called the limbic lobe/limbic system.

(159) Answer : (4)
Solution:
Brain stem include midbrain and hind brain except cerebellum.

(160) Answer : (1)
Solution:
Fibrous joints are present in cranium.

(161) Answer : (3)
Solution:
A centre present in pons called pneumotaxic centre can moderate the functions of the respiratory rhythm centre located in the medulla oblongata.

(162) Answer : (1)
Solution:

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Nerve impulses are strictly **unidirectional**, traveling from the **dendrites** (which receive signals), through the **cell body** (soma), and along the **axon** to the **axon terminals**.

(163) Answer : (3)
Solution:
Myasthenia gravis: Auto immune disorder affecting neuromuscular junction leading to fatigue, weakening and paralysis of skeletal muscle.

(164) Answer : (3)
Hint:
Function of cerebrum
Solution:
Hypothalamus is the part of the forebrain.
The hypothalamus contains a number of centres, which control body temperature, urge for eating and drinking. It also contains several groups of neurosecretory cells, which secrete hormones called hypothalamic hormones.

(165) Answer : (2)
Solution:
Unmyelinated nerve fibre of PNS is enclosed by a Schwann cell that does not form a myelin sheath around the axon, and is commonly found in autonomous and the somatic neural system.

(166) Answer : (4)
Solution:
The $Na^+ – K^+$ pump transports 3 Na^+ outwards for 2 K^+ into the axon maintaining the polarisation state when neuron is in resting state.

(167) Answer : (1)
Solution:
Skeletal system of humans consists of bones and a few cartilages.

(168) Answer : (3)
Solution:
Parietal and temporal are paired cranial bones while mandible is an unpaired facial bone.

(169) Answer : (2)
Solution:
Hyoid is U shaped. Sternum lack neural canal.

(170) Answer : (2)
Solution:
Coxal bone and patella are part of appendicular skeleton.

(171) Answer : (3)
Solution:
The bones of the hand (fore limb) are humerus, radius and ulna, carpals (wrist bones– 8 in number), metacarpals (palm bones – 5 in number) and phalanges (digits – 14 in number).

(172) Answer : (3)
Solution:
Pivot joint – Between atlas and axis vertebra
Gliding joint – Between carpals
Hinge joint – Between humerus in upper arm and ulna and radius in forearm

(173) Answer : (3)
Solution:
Bipolar neurons are found in the retinal layer of the eye.

(174) Answer : (2)
Solution:
Cerebral aqueduct is present in midbrain.

(175) Answer : (1)
Solution:
The process through which two or more organs interact and complement the functions of one another is called coordination.

(176) Answer : (4)
Solution:
Three major regions make up the brain stem; mid brain, pons and medulla oblongata. Brain stem forms the connections between the brain and spinal cord.

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(177) Answer : (4)
Solution:
Number of cranial bones = 8
Number of carpal bone in each wrist = 8

(178) Answer : (2)
Solution:
The mid brain is located between the thalamus/hypothalamus of the forebrain and pons of the hindbrain.

(179) Answer : (3)
Solution:
Sternum is present on mid-ventral part of thorax. Scapula is flat bone present on dorsolateral side of thorax.

(180) Answer : (1)
Solution:
Cerebellum is the part of the hind brain, while corpus callosum, association areas and hypothalamus are parts of forebrain.

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